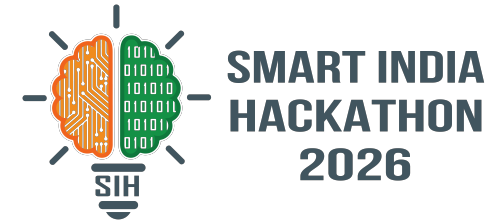
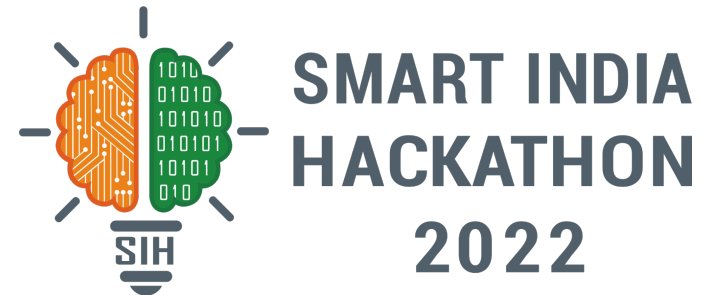


## TITLE PAGE



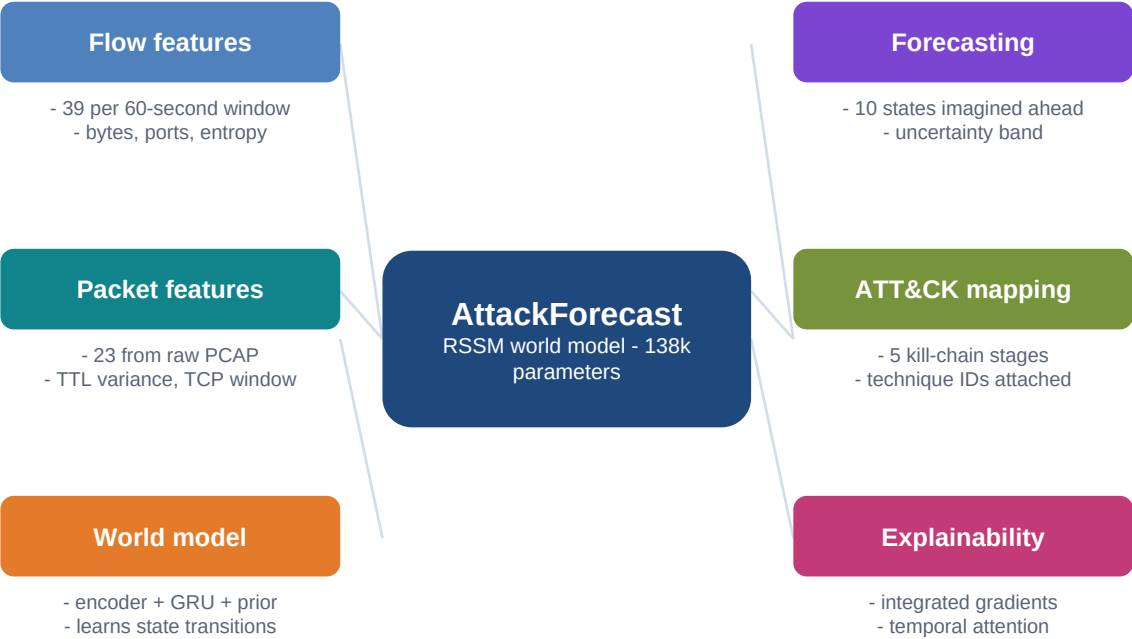
**Problem Statement ID - 26153**  
**Problem Statement Title - AI based Network Attack Forecasting from  
Network Traffic Data**  
**Theme - Blockchain & Cybersecurity**  
**PS Category - Software**  
**Organisation - National Technical Research Organisation (NTRO)**  
**Team ID - 261**  
**Team Name - Git with It**



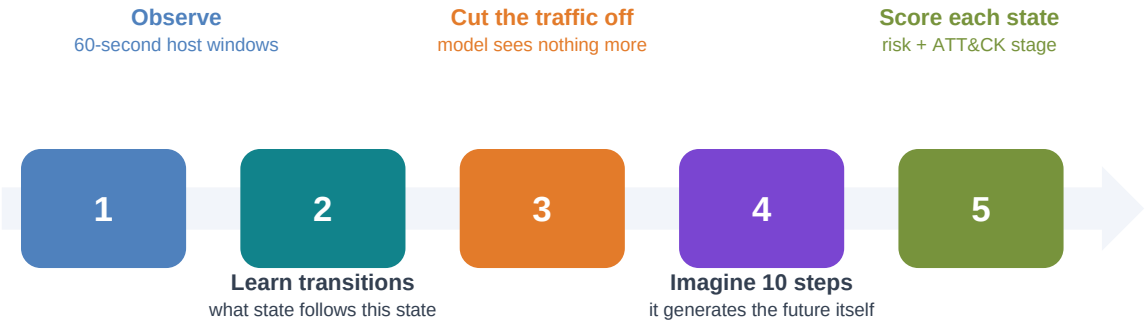


We do not classify traffic - we learn how a network's state evolves, then run that model forward to see where a host is heading.

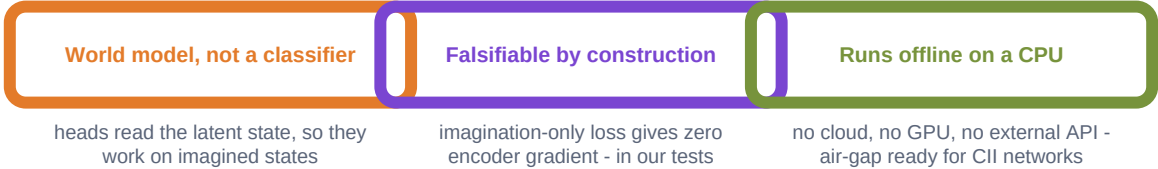
## AttackForecast - what it is made of



## How it addresses the problem

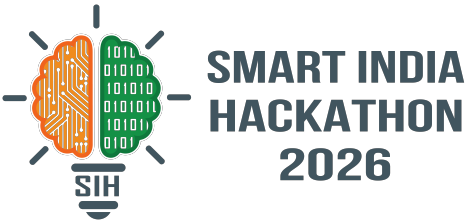


## Innovation and uniqueness



Working prototype today - 258,229 network states from all 13 CTU-13 captures - 28 of 30 infected hosts flagged at 4.9% false alarms

# TECHNICAL APPROACH



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## Architecture - traffic in, forecast out



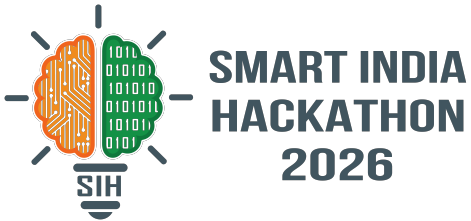
Validation: split by time, never randomly - baselines given identical features, scaler and labels - seeded rollouts, so the benchmark reproduces byte for byte

## Tech stack used

|                     |                                  |
|---------------------|----------------------------------|
| Language            | Python 3.13                      |
| Deep learning       | PyTorch (CPU-only)               |
| Packet parsing      | Scapy - raw PCAP, flow rebuild   |
| Data pipeline       | pandas + NumPy, vectorised       |
| Baselines & metrics | scikit-learn                     |
| Backend             | FastAPI + Uvicorn                |
| Frontend            | Single-file HTML, zero CDN       |
| Explainability      | Integrated Gradients + attention |
| Dataset             | CTU-13 - 13 real captures, CC BY |
| Deployment          | Fully offline, no external API   |



# FEASIBILITY AND VIABILITY



## Feasibility of AttackForecast

**Technical**  
CPU-only, 138k  
params

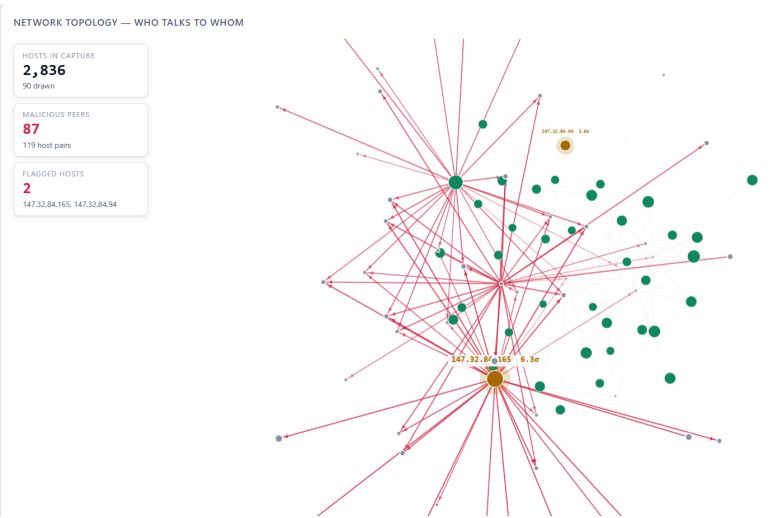
**Financial**  
zero licence cost

**Operational**  
consumes NetFlow  
already exported

**Legal**  
data never leaves the  
network

**Social**  
analyst keeps the  
decision

## Live output from the running prototype



3D topology - the compromised host and its malicious fan-out

## Potential challenges and risks

**Label leak**  
a metadata flag became a  
perfect label proxy

**Detection is only a tie**  
0.979 against 0.977 for  
logistic regression

**Short-burst  
compromise**  
the risk head needs sustained  
activity

**Stages do not cross  
families**  
progression did not transfer

## Strategies for overcoming these

**Found and removed**  
caught in our own audit; F1  
0.23 to 0.98

**Compete on  
forecasting**  
beats persistence at 9 of 10  
horizons

**Second channel**  
unsupervised surprise covers  
the gap

**Reported, not hidden**  
detection transfers;  
progression does not

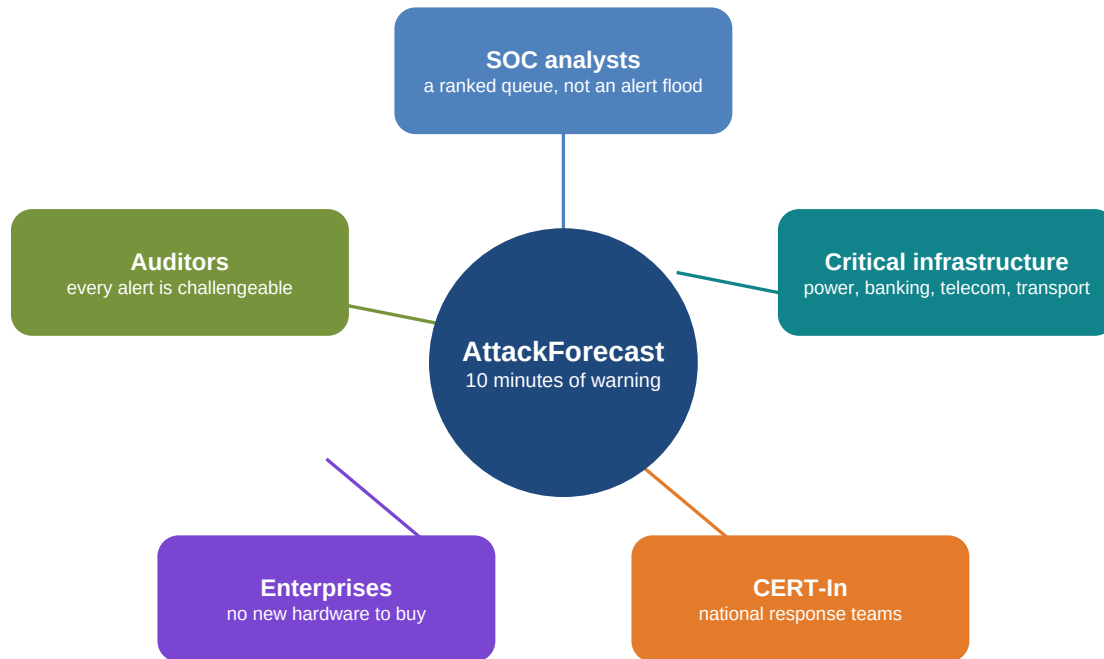
| Stage forecast macro-F1 by horizon |       |       |       |       |
|------------------------------------|-------|-------|-------|-------|
| Horizon                            | +2    | +4    | +6    | +10   |
| World model (ours)                 | 0.583 | 0.642 | 0.624 | 0.524 |
| "nothing changes" baseline         | 0.473 | 0.474 | 0.455 | 0.436 |

# IMPACT AND BENEFITS

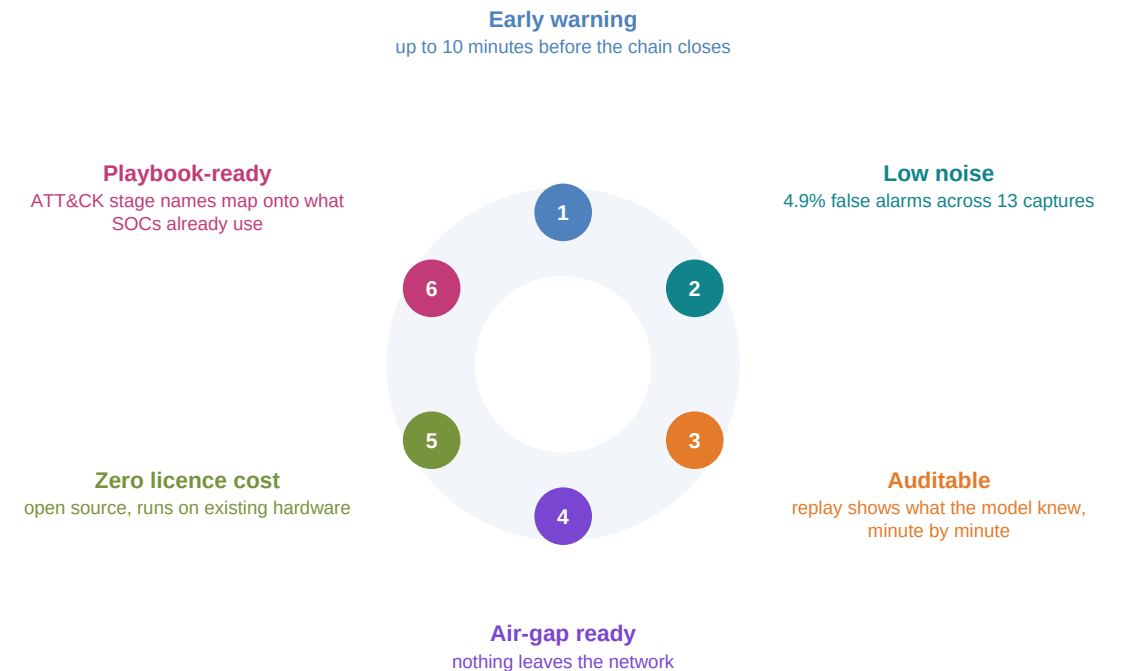
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Shifting the defender from reacting after compromise to acting during the kill chain

## Potential impact on the target audience



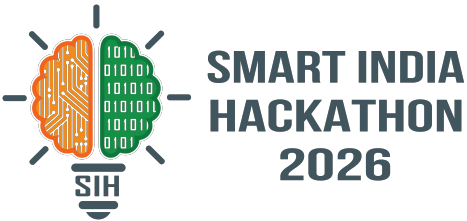
## Benefits of AttackForecast



Every claim on this deck is reproducible from the repository: one command prepares the data, one trains, one benchmarks.



# RESEARCH AND REFERENCES



## Datasets, standards and prior work



## Team - Git with It



Problem Statement 26153 (NTRO) - AI based Network Attack Forecasting from Network Traffic Data - Blockchain & Cybersecurity

Source, data preparation and benchmark: [github.com/HowSuyash/AttackForecast](https://github.com/HowSuyash/AttackForecast)